

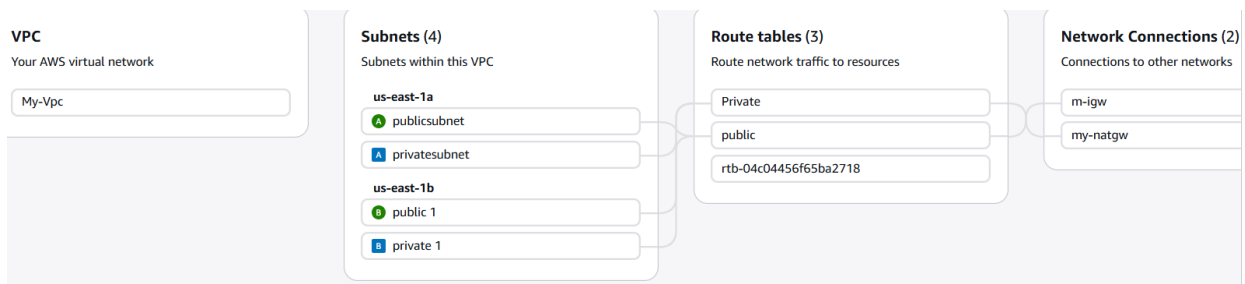
Task 6

Step No.	Task Category	Detailed Steps	Expected Outcome
1	Network Setup	Use existing VPC. Ensure 2 public subnets (ALB, Bastion) and 2 private subnets (EC2 in different AZ). Public subnet routes to IGW, private subnet routes to NAT Gateway.	Secure network setup completed
2	Security Group	Bastion SG: SSH (22) from your IP only. Private EC2 SG: SSH from Bastion SG and HTTP (80) from ALB SG. ALB SG: HTTP (80) from 0.0.0.0/0.	Proper security rules applied
3	Apache Setup	Launch private EC2 and add User Data to install Apache (httpd), start and enable service.	Apache running on private EC2
4	Website Content	Use instance metadata in User Data to display Instance ID and Availability Zone on index.html.	Dynamic page verified
5	Load Balancer	Create Application Load Balancer in public subnets. Create target group and register private EC2 instances.	Website accessible via ALB DNS
6	Auto Scaling	Create Launch Template with Apache User Data. Create Auto Scaling Group (Min 2, Desired 2, Max 4) and attach target group.	Auto scaling working correctly
7	Failover Test	Manually stop one EC2 instance and verify Auto Scaling launches a new instance automatically.	New EC2 launches automatically
8	Monitoring	Create CloudWatch alarms for CPU >70%, >80%, Memory >70%, >80%, Disk >70%, >80% and StatusCheckFailed.	Alarm triggers correctly
9	Notification	Create SNS topic, add email subscription, confirm email and attach SNS topic to alarms.	Email alert received
10	Cost Optimization	Create EventBridge rules to stop EC2/ASG at 7 PM and start at 9 AM automatically.	Automatic scheduling verified

Solution:

1. Network Setup (VPC & Subnets)

- Open **VPC Dashboard**
- Use an **existing VPC** (or create one if required).
- Create **4 subnets**:
- **2 Public Subnets** (different AZs) → for **ALB & Bastion**
- **2 Private Subnets** (different AZs) → for **EC2**
- Create and attach:
 - **Internet Gateway (IGW)** to VPC
 - **NAT Gateway** in one public subnet
- Route tables:
- Public route table → 0.0.0.0/0 → IGW
- Private route table → 0.0.0.0/0 → NAT Gateway
- Associate:
- Public subnets → public route table
- Private subnets → private route table



2. Security Groups

Bastion Security group

- Inbound:
 - SSH (22) - your Public IP only here
- Outbound:

- Allow all Traffic

ALB Security Group

- Inbound:
 - HTTP (80) - 0.0.0.0/0
- Outbound:
 - Allow all Traffic

Private Ec2 Security Group

- Inbound:
 - SSH (22) → Bastion SG
 - HTTP (80) → ALB SG
- Outbound:
 - Allow all Traffic

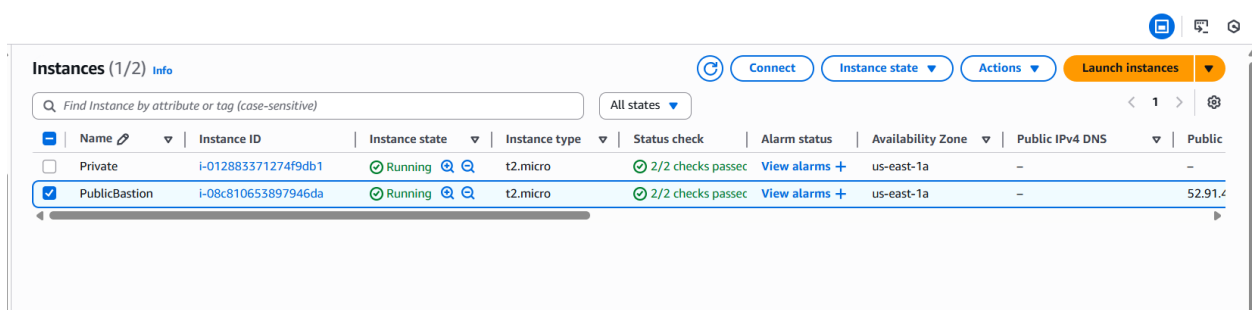
☐	Name	Security group ID	Security group name	VPC ID
☐	-	sg-08f59a544633d22ef	ALB-SG	vpc-01dab04e5b7d5eb7d
☐	-	sg-06785655b90612fd7	EC2-SG	vpc-01dab04e5b7d5eb7d

3. Apache Setup on private server

Bastion Server

EC2 – Launch Server

- AMI: Amazon Linux 2
- Subnet: public Subnet
- Security Group: bastion SG



☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
☐	Private	i-012883371274f9db1	Running	t2.micro	2/2 checks passed	View alarms	us-east-1a	-	-
☒	PublicBastion	i-08c810653897946da	Running	t2.micro	2/2 checks passed	View alarms	us-east-1a	52.91.4...	Public

Private server

EC2 – Launch Server

- AMI: Amazon Linux 2
- Subnet: private Subnet
- Security Group: private ec2 SG
- Key pair – create new

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

[Choose file](#)

```
#!/bin/bash
sudo yum update -y
sudo yum install httpd -y

TOKEN=$(curl -X PUT "http://169.254.169.254/latest/api/token" \
-H "X-aws-ec2-metadata-token-ttl-seconds: 21600")

INSTANCE_ID=$(curl -H "X-aws-ec2-metadata-token:$TOKEN" \
http://169.254.169.254/latest/meta-data/instance-id)

AZ=$(curl -H "X-aws-ec2-metadata-token:$TOKEN" \
http://169.254.169.254/latest/meta-data/placement/availability-zone)

sudo cat <<EOF > /var/www/html/index.html
<html>
```

☐ User data has already been base64 encoded

Instances (1/2) [Info](#)

Find Instance by attribute or tag (case-sensitive)

All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
☒	Private	i-012883371274f9db1	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	-	-
☐	PublicBastion	i-08c810653897946da	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	-	52.91.4

i-012883371274f9db1 (Private)

[Details](#) | [Status and alarms](#) | [Monitoring](#) | [Security](#) | [Networking](#) | [Storage](#) | [Tags](#)

Install apache on this private server

- Sudo yum update -y
- Sudo yum install httpd -y
- Sudo systemctl start httpd
- Sudo systemctl enable httpd

Connect bastion to private


```
#!/bin/bash
yum update -y
yum install httpd -y

TOKEN=$(curl -X PUT "http://169.254.169.254/latest/api/token" \
-H "X-aws-ec2-metadata-token-ttl-seconds: 21600")

INSTANCE_ID=$(curl -H "X-aws-ec2-metadata-token:$TOKEN" \
http://169.254.169.254/latest/meta-data/instance-id)

AZ=$(curl -H "X-aws-ec2-metadata-token:$TOKEN" \
http://169.254.169.254/latest/meta-data/placement/availability-zone)

cat <<EOF > /var/www/html/index.html
<html>
<body>
<h1>Welcome to Apache Server</h1>
<p>Instance ID: $INSTANCE_ID</p>
<p>Availability Zone: $AZ</p>
</body>
</html>
EOF
systemctl start httpd
systemctl enable httpd
```

5. Create A load balancer :

- Go to EC2 → Load Balancers
- Create Application Load Balancer
- Select:
 - Public subnets
 - ALB Security Group
- Create Target Group:
 - Type: Instance
 - Protocol: HTTP (80)
- Register private EC2 instances
- Attach target group to ALB listener

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)

Scheme can't be changed after the load balancer is created.

☒ Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type [Info](#)

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ IPv4

Includes only IPv4 addresses.

☐ Dualstack

Includes IPv4 and IPv6 addresses.

☐ Dualstack without public IPv4

Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with internet-facing load balancers only.

Network mapping [Info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC [Info](#)

You can optionally choose to configure an IPAM pool as the preferred source for your load balancer's IP addresses. Create or view pools in the [Amazon VPC IP Address Manager console](#).

☐ Use IPAM pool for public IPv4 addresses

The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets [Info](#)

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ us-east-1a (use1-az4)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0af133a5591a33c48
IPv4 subnet CIDR: 10.0.1.0/24

publicsubnet ▾

☒ us-east-1b (use1-az6)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-077ac8249131b6511
IPv4 subnet CIDR: 10.0.3.0/24

public1 ▾

Security groups [Info](#)

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups



ALB-SG

69-08f59b544633d22ef VPC: vpc-01dab04e5b7d5eb7d

Default action [Info](#)

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

☒ Forward to target groups☐ Redirect to URL☐ Return fixed responseForward to target group [Info](#)

Choose a target group and specify routing weight or [create target group](#).

Target group

newtask6

Target type: Instance, IPv4 | Target stickiness: Off

HTTP ▾



Weight

1

0-999

Percent

100%

+ Add target group

You can add up to 4 more target groups.

Target group stickiness [Info](#)

Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

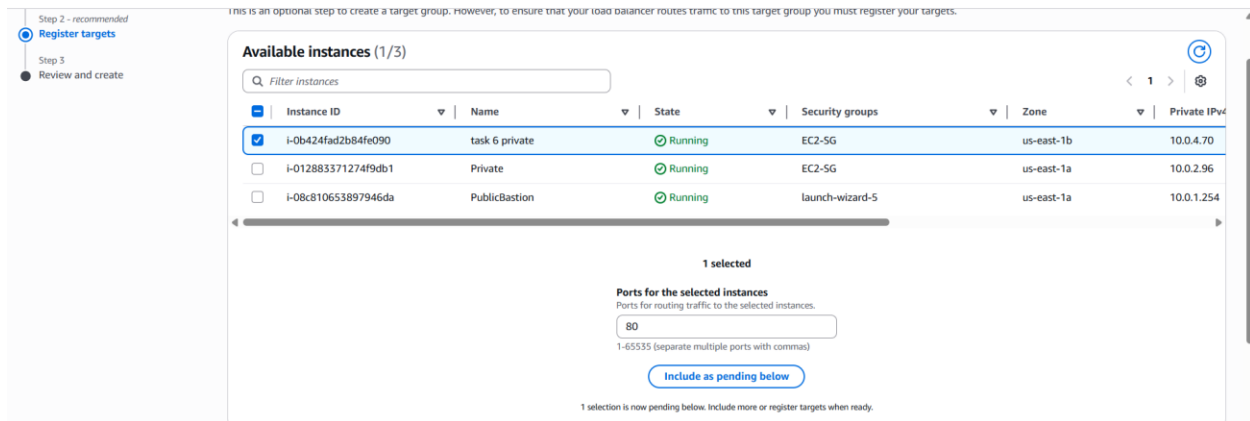
☐ Turn on target group stickiness

Listener tags - optional

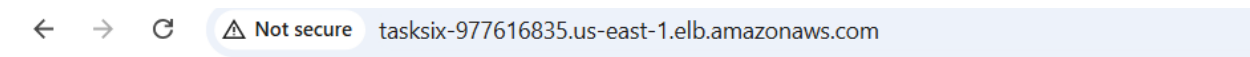
Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

Add listener tag

You can add up to 50 more tags.



Now website is accessible through ALB DNS



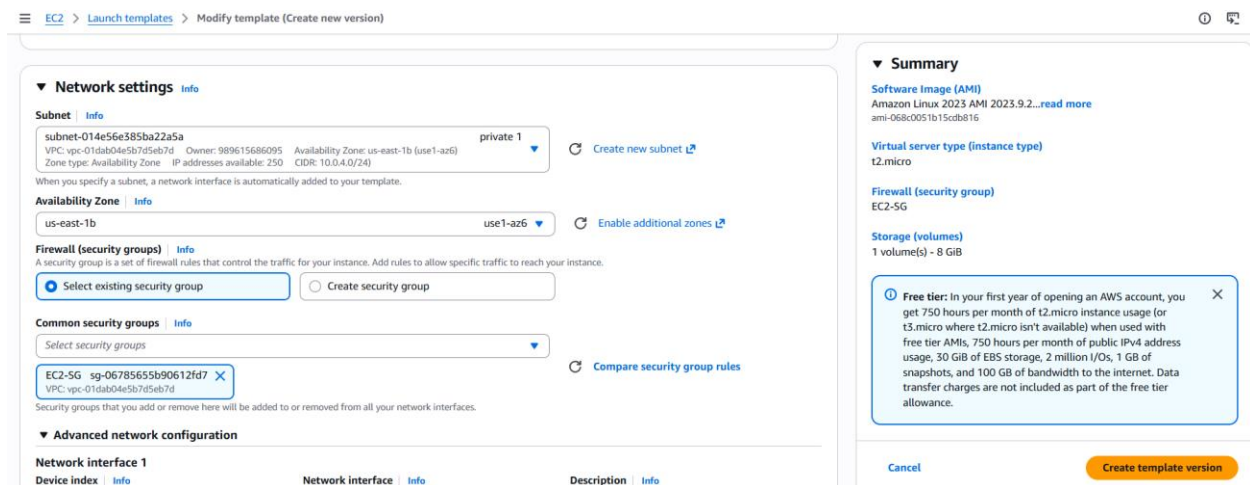
Welcome to Apache Server

Instance ID: i-0b424fad2b84fe090

Availability Zone: us-east-1b

6. Auto Scaling Group (ASG)

Create Launch template with same user data required for private ec2 to install apache on server



EC2

Launch templates

Modify template (Create new version)

Metadata response hop limit

2

Allow tags in metadata

Don't include in launch template

User data - optional

Choose file

```
#!/bin/bash
yum install httpd -y
TOKEN=" curl -X PUT "http://169.254.169.254/latest/api/token" \
-H "X-aws-ec2-metadata-token-ttl-seconds: 21600"
INSTANCE_ID=" curl -H "X-aws-ec2-metadata-token:$TOKEN" \
http://169.254.169.254/latest/meta-data/instance-id"
AZ=" curl -H "X-aws-ec2-metadata-token:$TOKEN" \
http://169.254.169.254/latest/meta-data/placement/availability-zone"

echo "<h1>Instance ID: $INSTANCE_ID</h1><h2>AZ: $AZ</h2>" > /var/www/html/index.html

systemctl start httpd
systemctl enable httpd
```

User data has already been base64 encoded

Summary

Software Image (AMI)

Amazon Linux 2023 AMI 2023.9.2...read more

ami-068c0051b15cdb816

Virtual server type (instance type)

t2.micro

Firewall (security group)

EC2-SG

Storage (volumes)

1 volume(s) - 8 GiB

Free tier:

In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance.

Cancel

Create template version

Auto Scaling Group

- Create ASG using launch template
- Subnets: Private subnets
- Desired: 2 | Min: 2 | Max: 4
- Attach to ALB target group

EC2

Auto Scaling groups

Create Auto Scaling group

Step 4 - optional

Configure group size and scaling

Step 5 - optional

Add notifications

Step 6 - optional

Add tags

Step 7

Review

new scaling group name

tasksix

Must be unique to this account in the current Region and no more than 255 characters.

Launch template

For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template

Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

Apache

Create a launch template

Version

Default (1)

Create a launch template version

Description

Apache

AMI ID

ami-068c0051b15cdb816

Key pair name

Titoo

Launch template

Apache

lt-0415bce12bc6f21ac

Security groups

-

Security group IDs

sg-0a99d9b5d1916ba17

Instance type

t2.micro

Request Spot Instances

No

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 6 - optional

Add tags

Step 7

Review

Instance type

t2.micro

Network

info

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC

Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-01dab04e5b7d5eb7d (My-Vpc)

10.0.0.0/16

Create a VPC

Availability Zones and subnets

Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

use1-azf (us-east-1b) | subnet-014e56e385ba22a5a (private 1)

10.0.4.0/24

Create a subnet

Availability Zone distribution - new

Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

Balanced best effort

If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.

Balanced only

If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy Availability Zone to preserve balanced distribution.

EC2 > Auto Scaling groups > Create Auto Scaling group

Add notifications

Step 6 - optional

Add tags

Step 7

Review

Desired capacity

Specify your group size.

2

Scaling

info

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

2

Equal or less than desired capacity

Max desired capacity

2

Equal or greater than desired capacity

Automatic scaling - optional

Choose whether to use a target tracking policy

Info

You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

No scaling policies

Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

Target tracking scaling policy

Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Instance maintenance policy

info

Control your Auto Scaling group's availability during instance replacement events. This includes health checks, instance refreshes, maximum instance lifetime features and events that happen automatically to keep your group balanced, called rebalancing events.

Choose a replacement behavior depending on your availability requirements

EC2 > Auto Scaling groups > Create Auto Scaling group

Instance scale-in protection

Disabled

Monitoring

Disabled

Default instance warmup

Disabled

Capacity Reservation preference

Preference

Default

Capacity Reservation IDs

-

Resource Groups

-

Step 5: Add notifications

Edit

Notifications

No notifications

Step 6: Add tags

Edit

Tags (0)

Key

Value

Tag new instances

No tags

Preview code

Cancel

Previous

Create Auto Scaling group

EC2 > Auto Scaling groups > tasksize > Edit

Load balancing - optional
Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

Load balancers

☒ Application, Network or Gateway Load Balancer target groups
Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.

Select target groups

newtasksize | HTTP
Application Load Balancer: tasksize

☐ Classic Load Balancers

Create and attach new load balancers

Add a new load balancer

7. Failover Test

- Goto Ec2- Instance
- Manually stop one instance.

Observer ASG will launch ec2 instance

Instances (5) Info

Last updated less than a minute ago

Connect Instance state Actions Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
☐	task 6 private	i-0b424fad2b84fe090	Running	t2.micro	2/2 checks passed	View alarms	us-east-1b	-	-
☐	Private	i-012883371274f9db1	Running	t2.micro	2/2 checks passed	View alarms	us-east-1a	-	-
☐		i-02fce7cf341637171	Running	t2.micro	2/2 checks passed	View alarms	us-east-1a	-	-
☐	PublicBastion	i-08c810653897946da	Running	t2.micro	2/2 checks passed	View alarms	us-east-1a	-	52.91.4
☐		i-0466833838753a8ca	Running	t2.micro	2/2 checks passed	View alarms	us-east-1b	-	-

8. Monitoring

Create alarms for:

- CPU Utilization > 70%
- CPU Utilization > 80%
- Memory Utilization > 70% & 80% (requires CloudWatch Agent)
- Disk Utilization > 70% & 80%
- StatusCheckFailed = 1

Go to CloudWatch – Alarms

CPU Utilization

CloudWatch > Alarms > Create alarm

Step 1: Specify metric and conditions

Instance name: task 6 private

Statistic: Average

Period: 5 minutes

Conditions

Threshold type: ☒ Static (Use a value as a threshold)

Whenever CPUUtilization is... Define the alarm condition:

☐ Greater > threshold

☒ Greater/Equal >= threshold

☐ Lower/Equal <= threshold

☐ Lower < threshold

than... Define the threshold value: 70

CloudWatch > Alarms > Create alarm

Alarm recommendations available
Turn on Recommendations to pre-populate the wizard with the recommended alarms.

Step 1: Specify metric and conditions

Step 2: Configure actions

Step 3: **Add alarm details**

Step 4: Preview and create

Add alarm details

Name and description

Alarm name: HighCPU Utilization

Alarm description - optional [View formatting guidelines](#)

[Edit](#) [Preview](#)

This is an H1
 double asterisks will produce strong character
 This is [an example](https://example.com/) inline link.

Up to 1024 characters (0/1024)

Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

CloudWatch > Alarms > Create alarm

Instance name: task 6 private

Statistic: Average

Period: 5 minutes

Conditions

Threshold type: Static

Whenever CPUUtilization is Greater/Equal (>=)

than... 70

[Additional configuration](#)

Step 2: Configure actions [Edit](#)

Follow same for 80% CPU utilization

- ❖ Memory Utilization(required CW agent)
- sudo yum install amazon-cloudwatch-agent -y

- `sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-config-wizard`
 - OS: Linux
 - Metrics: Memory, Disk
 - Namespace: CWAgent

`sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -a start`

```
[ec2-user@ip-10-0-4-70 ~]$ sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -a start
***** processing amazon-cloudwatch-agent *****
```

`sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -m ec2 -a status`

```
[ec2-user@ip-10-0-4-70 ~]$ sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -m ec2 -a status
{
  "status": "running",
  "starttime": "2025-12-24T04:42:44+00:00",
  "configstatus": "configured",
  "version": "1.300060.1"
}
[ec2-user@ip-10-0-4-70 ~]$
```

Add IAM role for CWAgent to select metrics.

- Go to IAM → Roles → Create role
- Select EC2 as the trusted entity
- Attach the CloudWatchAgentServerPolicy managed policy

Step 1: Select trusted entity

Select trusted entity

Trusted entity type

- ☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case
EC2

Choose a use case for the specified service.

Use case

- ☒ **EC2**
Allows EC2 instances to call AWS services on your behalf.
- ☐ **EC2 Role for AWS Systems Manager**
Allows EC2 instances to call AWS services like CloudWatch and Systems Manager on your behalf.

Step 2: Add permissions

☐		AmazonCloudWatchRUMReadOnlyAccess	AWS managed	Grants read only permissions for the A...
☐		AmazonDMSCloudWatchLogsRole	AWS managed	-
☐		AmazonGrafanaCloudWatchAccess	AWS managed	This policy grants access to Amazon Cl...
☐		AWSAppSyncPushToCloudWatchLogs	AWS managed	-
☐		CloudWatchActionsEC2Access	AWS managed	-
☐		CloudWatchAgentAdminPolicy	AWS managed	-
☒		CloudWatchAgentServerPolicy	AWS managed	-
☐		CloudWatchApplicationInsightsFullAccess	AWS managed	Provides full access to CloudWatch Ap...
☐		CloudWatchApplicationInsightsReadOnlyAccess	AWS managed	Provides read only access to CloudWat...
☐		CloudWatchApplicationSignalsFullAccess	AWS managed	Provide full access to CloudWatch App...
☐		CloudWatchApplicationSignalsReadOnlyAccess	AWS managed	Provides read only access to CloudWat...
☐		CloudWatchAutomaticDashboardsAccess	AWS managed	Provides access to the non-CloudWat...
☐		CloudWatchCrossAccountSharingConfiguration	AWS managed	Provides capabilities to manage Obser...
☐		CloudWatchEventsBuiltinTargetExecutionAccess	AWS managed	-
☐		CloudWatchEventsFullAccess	AWS managed	-
☐		CloudWatchEventsInvocationAccess	AWS managed	-

Now create Memory Usage Alaram

- Go to AWS Console → CloudWatch → Alarms → All alarms → Create alarms.
- Click Select metrics.
- Navigate to CWAgent

Memory Usage for 70%

CloudWatch > Alarms > Create alarm

Step 1
Specify metric and conditions

Step 2
Configure actions

Step 3
Add alarm details

Step 4
Preview and create

Specify metric and conditions

Metric

Graph

Preview of the metric or metric expression and the alarm threshold.

Select metric

Cancel

Next

Select metric

Untitled graph

1h3h12h1d3d1wCustom

UTC timezone

Line

1

0.5

0

02:4503:0003:1503:3003:4504:0004:1504:3004:4505:0005:1505:30

Your CloudWatch graph is empty.
Select some metrics to appear here.

Browse (1,539)

Multi source query

Graphed metrics

Options

Source

=

Add math

Add query

Alarm recommendations

Graph with SQL

Graph search

Search for any metric, dimension, resource id or account id

Custom namespaces

CWAgent8

Select metric

mem_used_percent

1h3h12h1d3d1wCustom

UTC timezone

Line

Percent

19.2

18.8

18.4

02:4503:0003:1503:3003:4504:0004:1504:3004:4505:0005:1505:30

mem_used_percent

Browse (1)

Multi source query

Graphed metrics (1)

Options

Source

=

Add math

Add query

Alarm recommendations

Graph with SQL

Graph search

Search for any metric, dimension, resource id or account id

InstancedId

< 1 >

Instance name 1/1

InstanceId

Metric name

Alarms

task 6 private

i-0b424fad2b84fe...

mem_used_percent

No alarms

CloudWatch > Alarms > Create alarm

Q Average X

Period
1 minute

Conditions

Threshold type

☒ Static
Use a value as a threshold

☐ Anomaly detection
Use a band as a threshold

Whenever mem_used_percent is...

Define the alarm condition.

☒ Greater
> threshold

☐ Greater/Equal
≥ threshold

☐ Lower/Equal
≤ threshold

☐ Lower
< threshold

than...

Define the threshold value.

70

Must be a number.

▼ Additional configuration

Datapoints to alarm

Define the number of datapoints within the evaluation period that must be breaching to cause the alarm to go to ALARM state.

CloudWatch > Alarms > Create alarm

○ Add alarm details
Step 4
○ Preview and create

Define the alarm state that will trigger this action.

☒ In alarm
The metric or expression is outside of the defined threshold.

☐ OK
The metric or expression is within the defined threshold.

☐ Insufficient data
The alarm has just started or not enough data is available.

Send a notification to the following SNS topic

Define the SNS (Simple Notification Service) topic that will receive the notification.

☒ Select an existing SNS topic

☐ Create new topic

☐ Use topic ARN to notify other accounts

Send a notification to...

Q Default_CloudWatch_Alarms_Topic X

Only topics belonging to this account are listed here. All persons and applications subscribed to the selected topic will receive notifications.

Email (endpoints)
ajitjadhavaj07@gmail.com - View in SNS Console

Add notification

Lambda action

Add Lambda action

Follow same for Memory Usage Above 80%

Disk Usage:

For above 70%

CloudWatch > Alarms > Create alarm

Step 1
Specify metric and conditions
Step 2
○ Configure actions
Step 3
○ Add alarm details
Step 4
○ Preview and create

Metric

Graph
Preview of the metric or metric expression and the alarm threshold.

Select metric

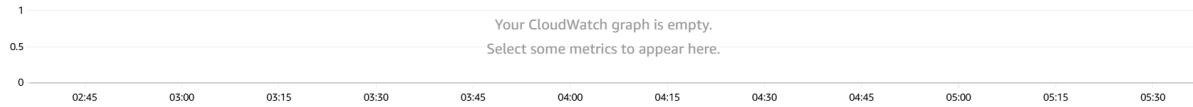
Cancel Next

Select metric



Untitled graph

1h 3h 12h 1d 3d 1w Custom UTC timezone Line



Browse (1,539) | Multi source query | Graphed metrics | Options | Source = **Add math** **Add query**

Search for any metric, dimension, resource id or account id

▼ Custom namespaces

CWAgent 8

Select metric



disk_used_percent

1h 3h 12h 1d 3d 1w Custom UTC timezone Line



Browse (7) | Multi source query | Graphed metrics (1) | Options | Source = **Add math** **Add query**

All > CWAgent > InstanceId, device, fstype, path

Search for any metric, dimension, resource id or account id

	Instance name 7/7	InstanceId	device	fstype	path	Metric name	Alarms
☒	task 6 private	i-0b424fad2b84fe...	xvda1	xfs	/	disk_used_percent	No alarms
☐	task 6 private	i-0b424fad2b84fe...	xvda128	vfat	/boot/efi	disk_used_percent	No alarms
☐	task 6 private	i-0b424fad2b84fe...	devtmpfs	devt....	/dev	disk_used_percent	No alarms
☐	task 6 private	i-0b424fad2b84fe...	tmpfs	tmpfs	/dev/shm	disk_used_percent	No alarms
☐	task 6 private	i-0b424fad2b84fe...	tmpfs	tmpfs	/run	disk_used_percent	No alarms

Cancel **Select metric**

CloudWatch > Alarms > Create alarm



Instance name
task 6 private

Statistic
Average

Period
1 minute

Conditions

Threshold type

☒ Static
Use a value as a threshold

☐ Anomaly detection
Use a band as a threshold

Whenever disk_used_percent is...

☐ Greater
> threshold

☒ Greater/Equal
≥ threshold

☐ Lower/Equal
≤ threshold

☐ Lower
< threshold

than...

Define the threshold value.

70

Must be a number.

CloudWatch > Alarms > Create alarm

Step 1
Specify metric and conditions

Step 2
Configure actions

Step 3
Add alarm details

Step 4
Preview and create

Add alarm details

Name and description

Alarm name

DiskUsage_70

Alarm description - optional [View formatting guidelines](#)

Edit

Preview

This is an H1
double asterisks will produce strong character
This is [an example](https://example.com/) inline link.

Up to 1024 characters (0/1024)

Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

Tags - optional [Info](#)

No tags associated with the resource.

Add new tag

Follow same for Disk Usage above 80%

For Service Check Fails

CloudWatch > Alarms > Create alarm

Step 1
Specify metric and conditions

Step 2
Configure actions

Step 3
Add alarm details

Step 4
Preview and create

Specify metric and conditions

Metric

Graph

Preview of the metric or metric expression and the alarm threshold.

Select metric

Cancel **Next**

Select metric

disk_used_percent

1h 3h 12h 1d 3d 1w Custom UTC timezone Line

1

0.5

0

03:00 03:15 03:30 03:45 04:00 04:15 04:30 04:45 05:00 05:15 05:30 05:45

Your CloudWatch graph is empty.
Select some metrics to appear here.

Browse (406)

Multi source query

Graphed metrics

Options

Source

Add math

Add query

EC2

Alarm recommendations

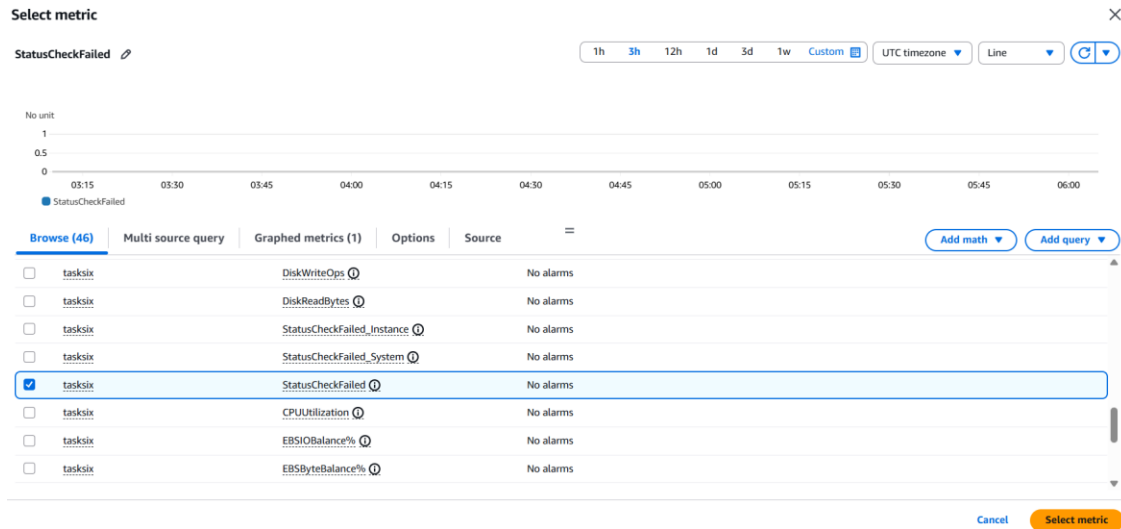
Graph with SQL

Graph search

Search for any metric, dimension, resource id or account id

By Auto Scaling Group 46

Per-Instance Metrics 360



CloudWatch > Alarms > Create alarm

tasksix

Statistic
Average

Period
1 minute

Conditions

Threshold type

Static
Use a value as a threshold

Anomaly detection
Use a band as a threshold

Whenever StatusCheckFailed is...

Define the alarm condition.

Greater
> threshold

Greater/Equal
>= threshold

Lower/Equal
≤ threshold

Lower
< threshold

than...

Define the threshold value.

1

Must be a number.

Additional configuration

Crate Alarm .

CloudWatch

Successfully created alarm SystemFail_Alarm. View alarm

Alarms (7)

Hide Auto Scaling alarms Clear selection Create composite alarm Actions Create alarm

Search Alarm state: Any Alarm type: Any Actions status: Any

	Name	State	Last state update (UTC)	Conditions	Actions
☐	SystemFail_Alarm	Insufficient data	2025-12-24 06:13:36	StatusCheckFailed >= 1 for 1 datapoints within 1 minute	Actions enabled
☐	DiskUsage_80	OK	2025-12-24 05:54:30	disk_used_percent >= 80 for 1 datapoints within 1 minute	Actions enabled
☐	DiskUsage_70	OK	2025-12-24 05:52:52	disk_used_percent >= 70 for 1 datapoints within 1 minute	Actions enabled
☐	MemoryUsage_80	OK	2025-12-24 05:43:14	mem_used_percent > 80 for 1 datapoints within 1 minute	Actions enabled
☐	memoryusage_70	OK	2025-12-24 05:42:06	mem_used_percent > 70 for 1 datapoints within 1 minute	Actions enabled
☐	CPU Usage above 80	OK	2025-12-24 04:14:04	CPUUtilization >= 80 for 1 datapoints within 5 minutes	Actions enabled
☐	HighCPU Utilization	OK	2025-12-24 04:13:50	CPUUtilization >= 70 for 1 datapoints within 5 minutes	Actions enabled

9. Notifications (SNS)

- Search SNS
- Navigate to Amazon SNS service.
- Click Topics → Create topic

Amazon SNS

Topics

Create topic

Details

Type [Info](#)

Topic type cannot be modified after topic is created

☐ FIFO (first-in, first-out)

- Strictly-preserved message ordering
- Exactly-once message delivery
- Subscription protocols: SQS

☒ Standard

- Best-effort message ordering
- At-least once message delivery
- Subscription protocols: SQS, Lambda, Data Firehose, HTTP, SMS, email, mobile application endpoints

Name

Default CloudAlarm

Maximum 256 characters. Can include alphanumeric characters, hyphens (-) and underscores (_).

Display name - optional [Info](#)

To use this topic with SMS subscriptions, enter a display name. Only the first 10 characters are displayed in an SMS message.

My Topic

Maximum 100 characters.

► Encryption - optional

Amazon SNS provides in-transit encryption by default. Enabling server-side encryption adds at-rest encryption to your topic.

► Access policy - optional [Info](#)

Amazon SNS

Subscriptions

Create subscription

Details

Topic ARN

arn:aws:sns:us-east-1:989615686095:Default_CloudWatch_Alarms_Topic

Protocol

The type of endpoint to subscribe

Email

Endpoint

An email address that can receive notifications from Amazon SNS.

test@example.com

After your subscription is created, you must confirm it. [Info](#)

► Subscription filter policy - optional [Info](#)

This policy filters the messages that a subscriber receives.

► Redrive policy (dead-letter queue) - optional [Info](#)

Send undeliverable messages to a dead-letter queue.

Cancel

Create subscription

To check it's work.

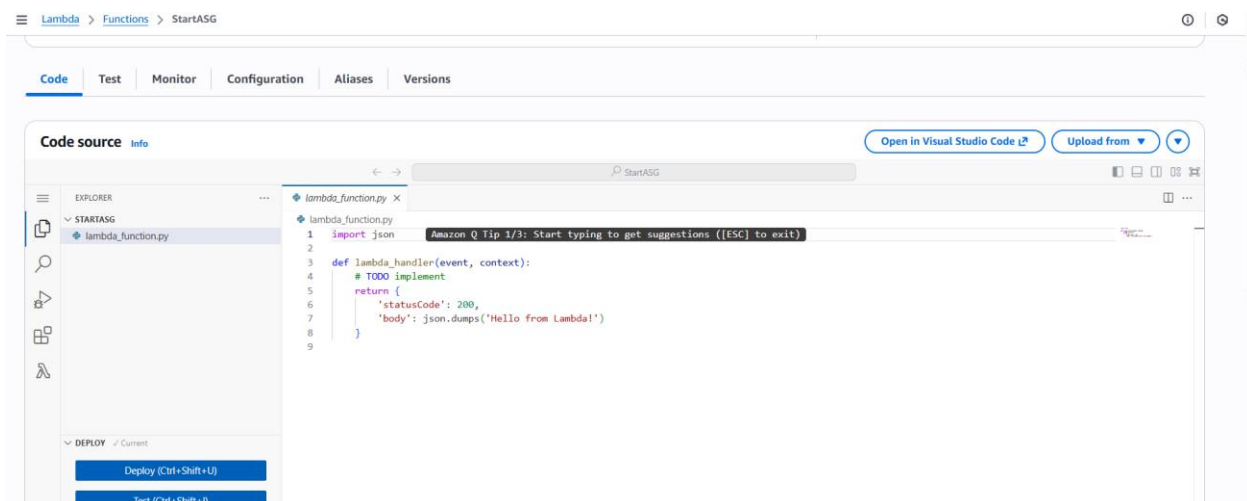
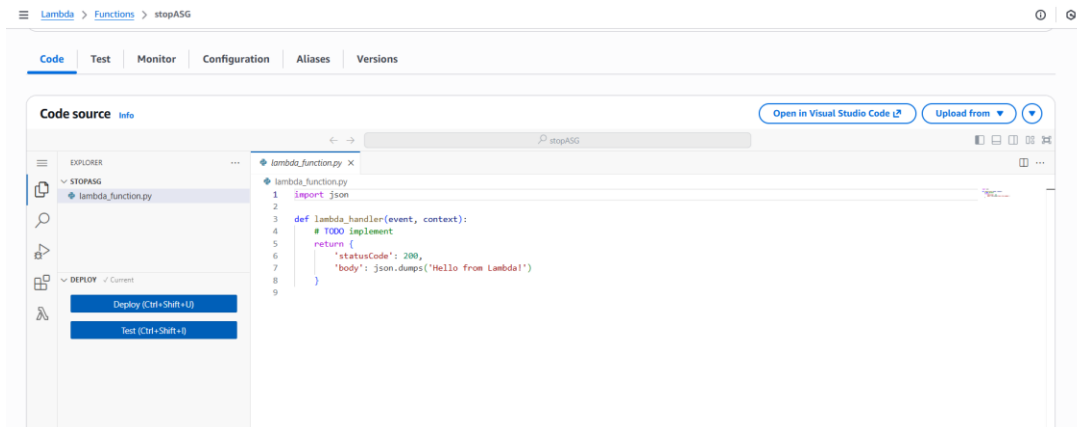
- Simulate high CPU load manually
- Verify that the email notification

10. Cost Optimization

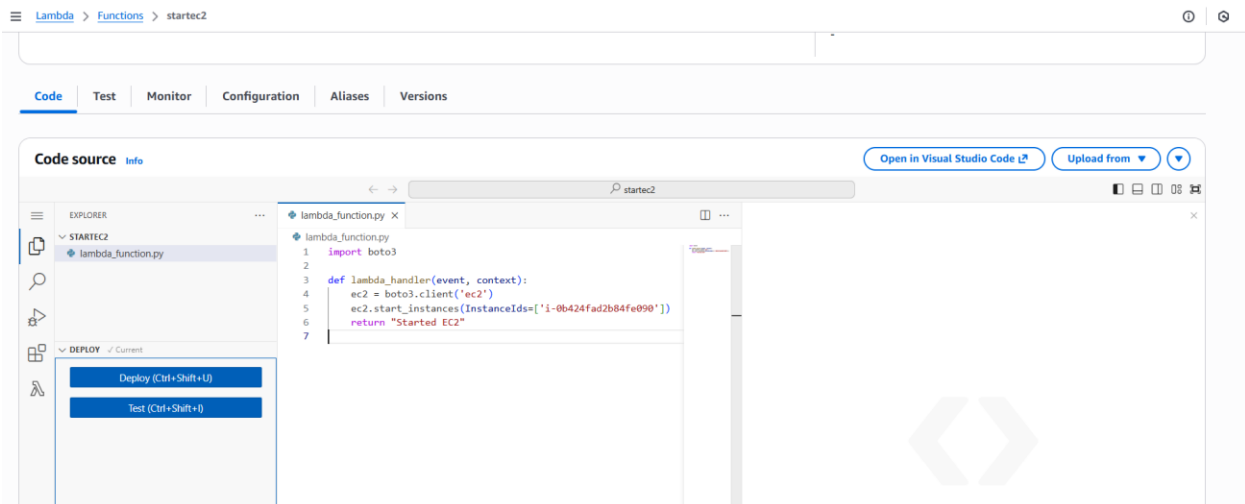
- Goto AWS - Navigate to Amazon EventBridge:
- Go to EventBridge service.

- Click Rules - Create rule.

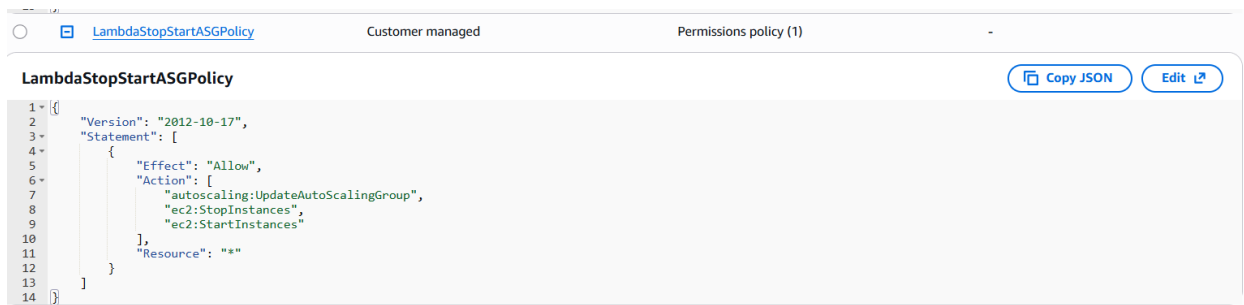
First create lambada functions to stop and start ASG



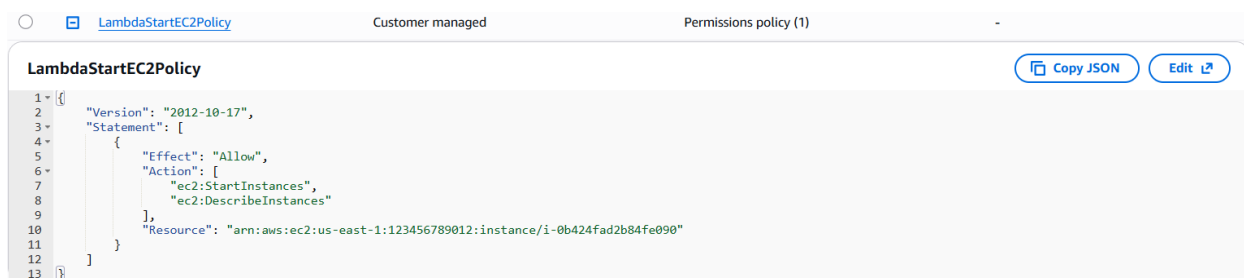
To start ec2 we also create a Lamda function for it



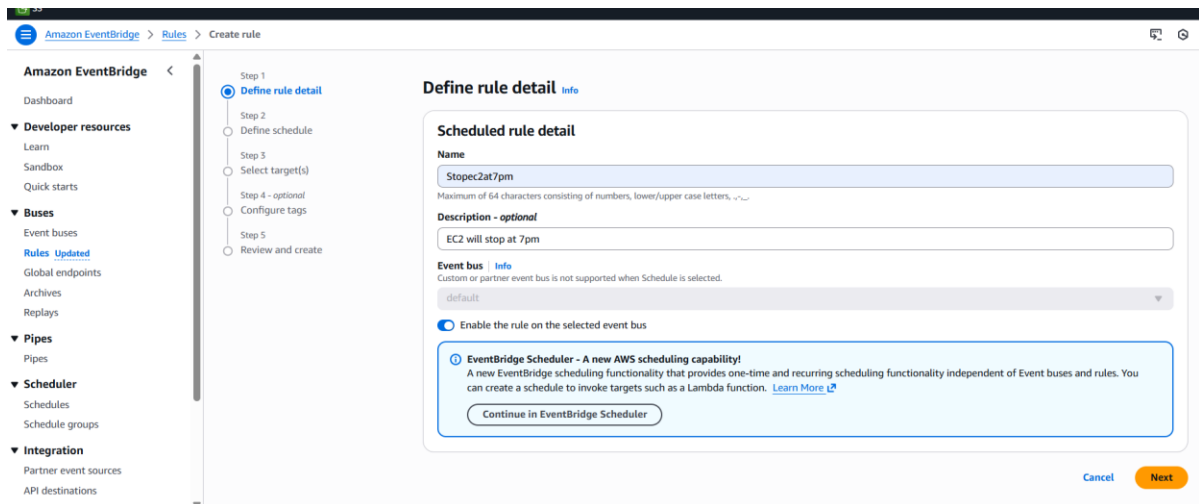
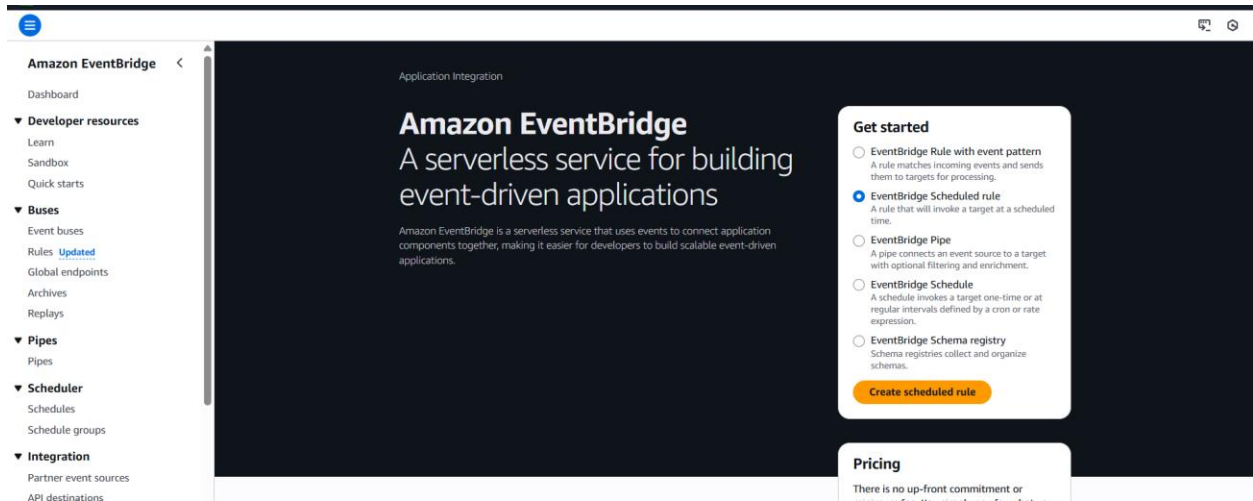
Attach start stop policies to IAM roles that have initially created by Lamda Functions



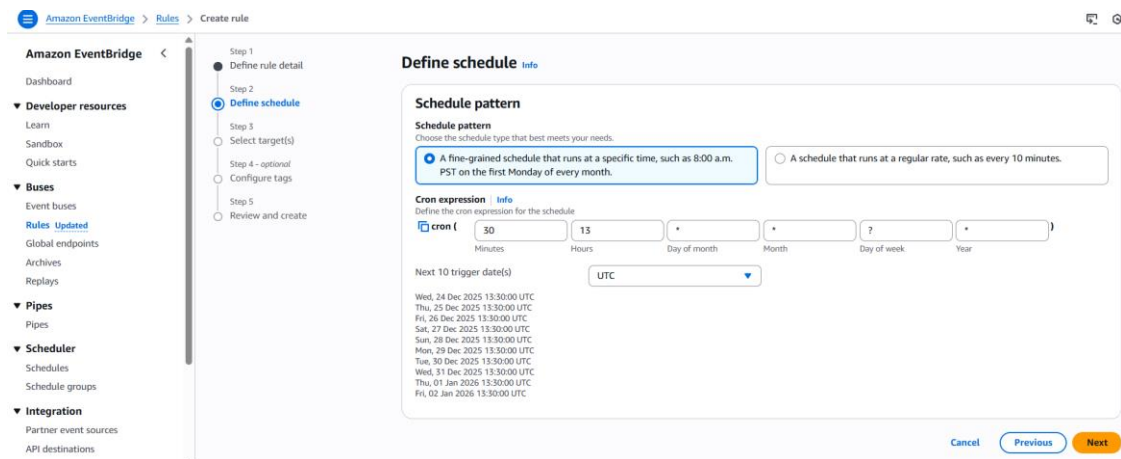
For EC2



Now create rule



To stop at 7 pm do this cron job



Amazon EventBridge > Rules > Create rule

Step 4 - optional
Configure tags

Step 5
Review and create

Amazon EventBridge

Dashboard

▼ **Developer resources**

Learn

Sandbox

Quick starts

▼ **Buses**

Event buses

[Rules Updated](#)

Global endpoints

Archives

Replays

▼ **Pipes**

Pipes

▼ **Scheduler**

Schedules

Schedule groups

▼ **Integration**

Partner event sources

API destinations

Target 1

Target types

Select an EventBridge event bus, EventBridge API destination (SaaS partner), or another AWS service as a target.

☐ EventBridge event bus

☐ EventBridge API destination

☒ AWS service

Select a target [Info](#)

Select target(s) to invoke when an event matches your event pattern or when schedule is triggered (limit of 5 targets per rule)

EC2 StopInstances API call

Instance ID

i-0b424fad2b84fe090

Execution role

EventBridge needs permission to send events to the target specified above. By continuing, you are allowing us to do so. [EventBridge and AWS Identity and Access Management](#)

☒ Create a new role for this specific resource

☐ Use existing role

Role name

Amazon_EventBridge_Invoke_Action_On_EC2_Instance_1286892488

► **Additional settings**

[Add another target](#) [Cancel](#) [Skip to Review and create](#) [Previous](#) [Next](#)

Select each function we had created for .

Amazon EventBridge > Rules > Create rule

Step 4 - optional
Configure tags

Step 5
Review and create

Amazon EventBridge

Dashboard

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Event buses

[Rules Updated](#)

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Partner event sources

API destinations

Target 1

Target types

Select an EventBridge event bus, EventBridge API destination (SaaS partner), or another AWS service as a target.

☐ EventBridge event bus

☐ EventBridge API destination

☒ AWS service

Select a target [Info](#)

Select target(s) to invoke when an event matches your event pattern or when schedule is triggered (limit of 5 targets per rule)

Lambda function

Target location

☒ Target in this account

☐ Target in another AWS account

Function

stopASG

► **Configure version/alias**

Permissions

☒ Use execution role (recommended)

Execution role

EventBridge needs permission to send events to the target specified above. By continuing, you are allowing us to do so. [EventBridge and AWS Identity and Access Management](#)

☒ Create a new role for this specific resource

☐ Use existing role

Role name

Amazon_EventBridge_Invoke_Lambda_1545672639

► **Additional settings**

Amazon EventBridge > Rules

Rule Startec2at7am was created successfully

Rules

A rule watches for specific types of events. When a matching event occurs, the event is routed to the targets associated with the rule. A rule can be associated with one or more targets.

Select event bus

Event bus
Select or enter event bus name

default

Rules on default event bus (2)

Find rules

Any status

Create rule

☐	Name	Status	Type	Event bus	ARN	Description
☐	Startec2at7am	Enabled	Scheduled Standard	default	arn:aws:events:us-east-1:989615686095:rule/Startec2at7am	-
☐	Stopce2at7pm	Enabled	Scheduled Standard	default	arn:aws:events:us-east-1:989615686095:rule/Stopce2at7pm	EC2 will stop at 7pm

Do same for start ec2 at 7 AM, use cron job (30 1 * * ? *)

Through this Automatic scheduling verified

EC2 > Instances

Instances (1/3)

Find Instance by attribute or tag (case-sensitive)

All states

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
☒	task 6 private	i-0b424fad2b84fe090	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1b	-	-
☐	Private	i-012883371274f9db1	Stopped	t2.micro	-	View alarms +	us-east-1a	-	-
☐	PublicBastion	i-08c810653897946da	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	-	3.88.10

i-0b424fad2b84fe090 (task 6 private)

CPU utilization (%)

Percent

3.02

1.51

0

07.45 08.00 08.15 08.30

Network in (bytes)

Bytes

15.0k

7.48k

0

07.45 08.00 08.15 08.30

Network out (bytes)

Bytes

12.2k

6.10k

0

07.45 08.00 08.15 08.30

Network packets in (count)

Count

63.8

31.9

0

07.45 08.00 08.15 08.30

Network packets out (count)

Count

54.8

27.4

07.45 08.00 08.15 08.30

Metadata no token (count)

No unit

1

07.45 08.00 08.15 08.30

CPU credit usage (count)

Count

0.039

07.45 08.00 08.15 08.30

CPU credit balance (count)

Count

53.5

07.45 08.00 08.15 08.30